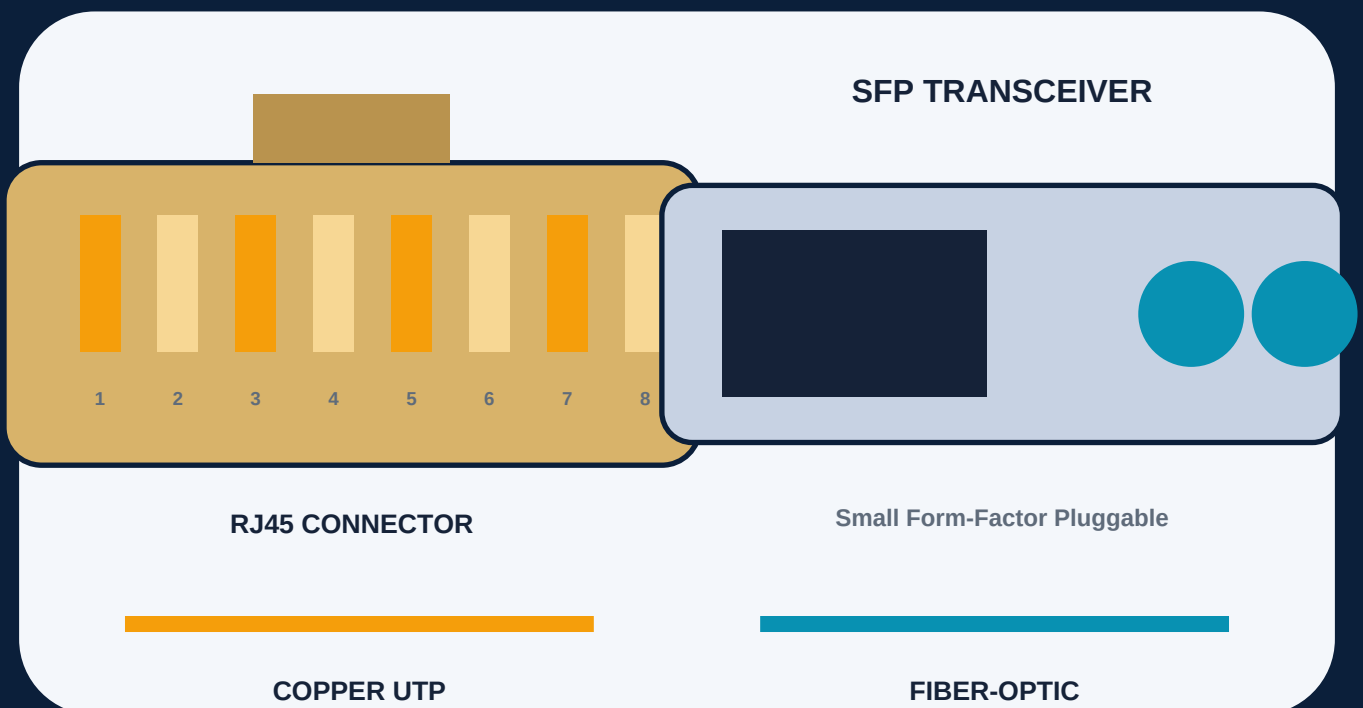


CCNA 200-301

Visual Flashcards

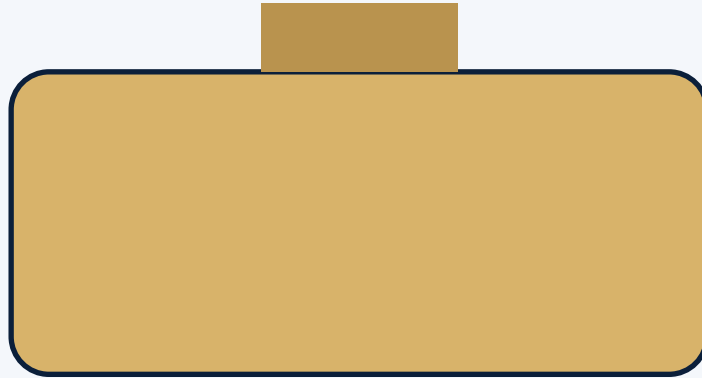
Day 02 - Interfaces and Cables

57 cards | Ethernet | UTP | RJ45 | copper | fiber | SFP



QUESTION

What is the name of the connector on the end of an Ethernet UTP cable?



RJ45 CONNECTOR

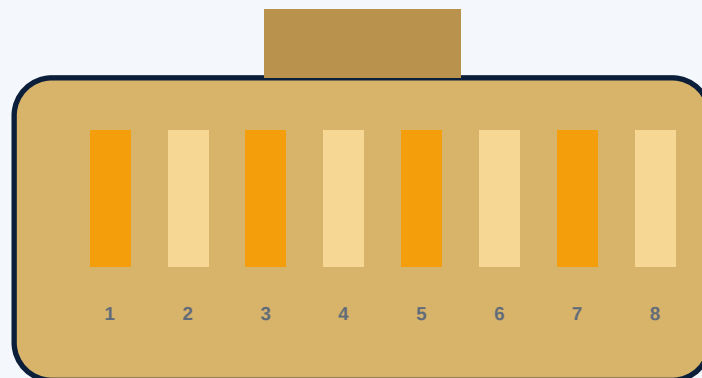
ANSWER

RJ45

RJ45 is the common connector used on Ethernet UTP patch cables.

QUESTION

How many pins are in an RJ45 connector?



RJ45 CONNECTOR

ANSWER

8

An RJ45 connector has eight pin positions.

QUESTION

How many bits are in 1 byte?

1 BYTE



8 bits

ANSWER

8

A byte contains eight bits.

QUESTION

Network speeds are measured in [...] per second.

measured in bits per second

NETWORK SPEED



8 bits

ANSWER

bits

Ethernet link rates are normally stated in bits per second.

QUESTION

How many bits are in 1 kilobit?

1 kilobit

=

1,000 bits

ANSWER

1,000 (one thousand)

Networking uses decimal SI rates: kilo = 1,000.

QUESTION

How many bits are in 1 megabit?

1 megabit

=

1,000,000 bits

ANSWER

1,000,000 (1 million)

Mega = one million bits.

QUESTION

How many bits are in 1 gigabit?

1 gigabit

=

1,000,000,000 bits

ANSWER

1,000,000,000 (1 billion)

Giga = one billion bits.

QUESTION

How many bits are in 1 terabit?

1 terabit

=

1,000,000,000,000 bits

ANSWER

1,000,000,000,000 (1 trillion)

Tera = one trillion bits.

QUESTION

Ethernet was originally defined in the IEEE [...] standard in 1983.

IEEE

802.3

IEEE ETHERNET STANDARD

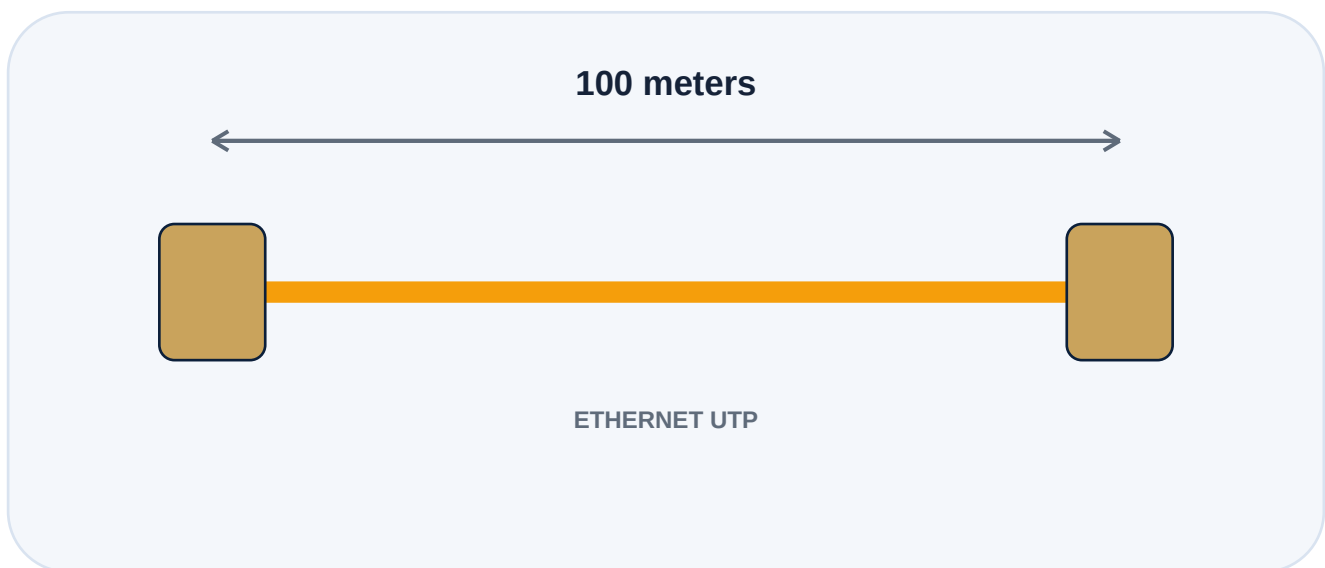
ANSWER

802.3

IEEE 802.3 is the Ethernet standards family.

QUESTION

What is the maximum length of an Ethernet UTP cable?



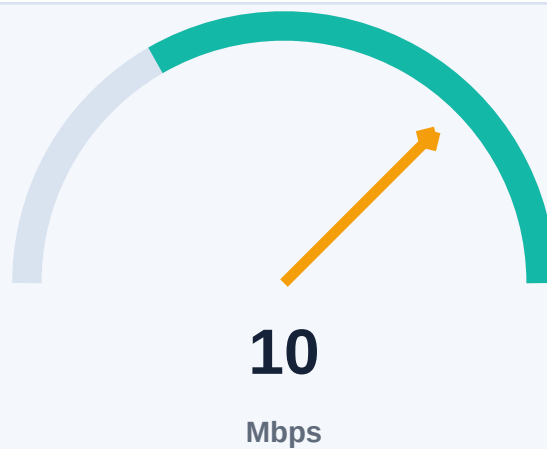
ANSWER

100 meters

Copper Ethernet UTP links are normally limited to 100 meters.

QUESTION

What is the speed of an 'Ethernet' connection?



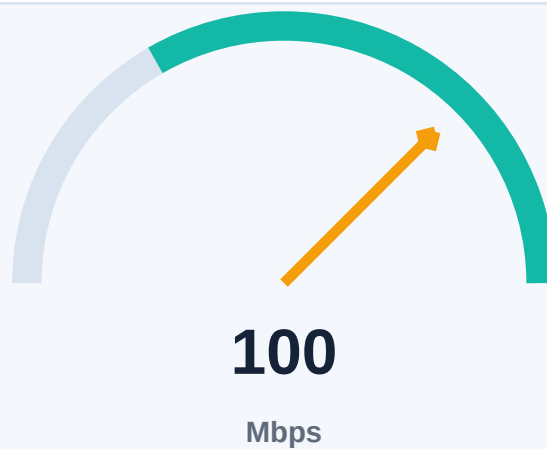
ANSWER

10 Mbps

10 Mbps is legacy Ethernet.

QUESTION

What is the speed of a 'FastEthernet' connection?



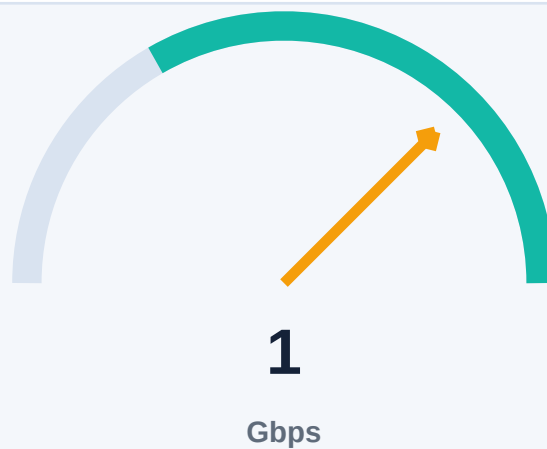
ANSWER

100 Mbps

FastEthernet = 100 Mbps.

QUESTION

What is the speed of a 'Gigabit Ethernet' connection?



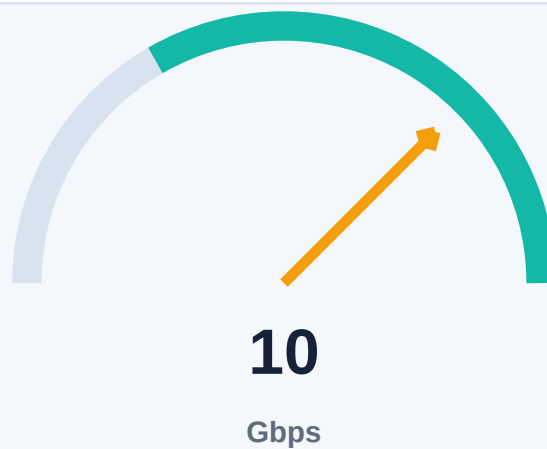
ANSWER

1 Gbps

Gigabit Ethernet = 1 Gbps.

QUESTION

What is the speed of a '10 Gig Ethernet' connection?



ANSWER

10 Gbps

10 Gigabit Ethernet = 10 Gbps.

QUESTION

IEEE standard [...] = 10 Mbps Ethernet (10BASE-T)

IEEE

802.3i

10BASE-T / 10 Mbps

ANSWER

802.3i

802.3i defines 10BASE-T.

QUESTION

IEEE standard [...] = 100 Mbps Ethernet (100BASE-T)

IEEE

802.3u

100BASE-T / 100 Mbps

ANSWER

802.3u

802.3u defines 100BASE-T.

QUESTION

IEEE standard [...] = 1 Gbps Ethernet (1000BASE-T)

IEEE

802.3ab

1000BASE-T / 1 Gbps

ANSWER

802.3ab

802.3ab defines 1000BASE-T.

QUESTION

IEEE standard [...] = 10 Gbps Ethernet (10GBASE-T)

IEEE

802.3an

10GBASE-T / 10 Gbps

ANSWER

802.3an

802.3an defines 10GBASE-T.

QUESTION

What does UTP stand for?

TWISTED PAIRS



ANSWER

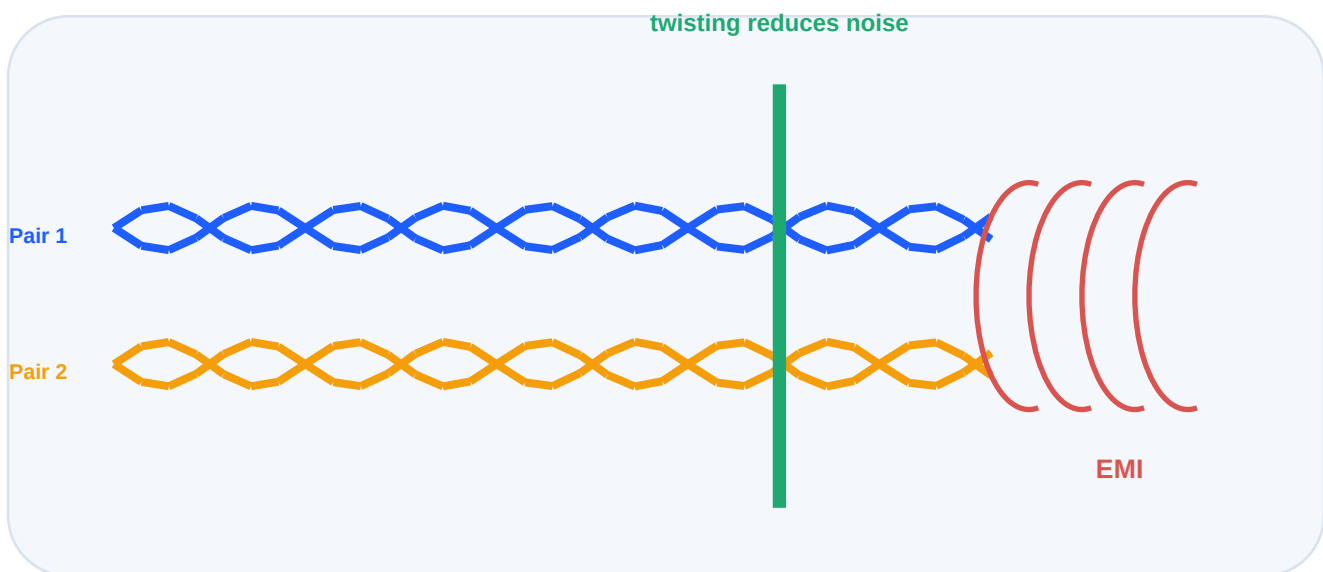
Unshielded Twisted Pair

UTP means Unshielded Twisted Pair.

QUESTION

Why are the wire pairs in a UTP cable twisted together?

TWISTED PAIRS



ANSWER

It protects against EMI (Electromagnetic Interference)

Twisting helps cancel electromagnetic interference.

QUESTION

How many pairs of wires are used in a 10BASE-T cable?

TWISTED PAIRS



2 pairs / 4 wires

ANSWER

2 (=4 wires)

10BASE-T uses two wire pairs.

QUESTION

How many pairs of wires are used in a 100BASE-T cable?

TWISTED PAIRS



2 pairs / 4 wires

ANSWER

2 (=4 wires)

100BASE-T uses two wire pairs.

QUESTION

How many pairs of wires are used in a 1000BASE-T cable?

TWISTED PAIRS



4 pairs / 8 wires

ANSWER

4 (=8 wires)

1000BASE-T uses all four pairs.

QUESTION

How many pairs of wires are used in a 10GBASE-T cable?

TWISTED PAIRS



4 pairs / 8 wires

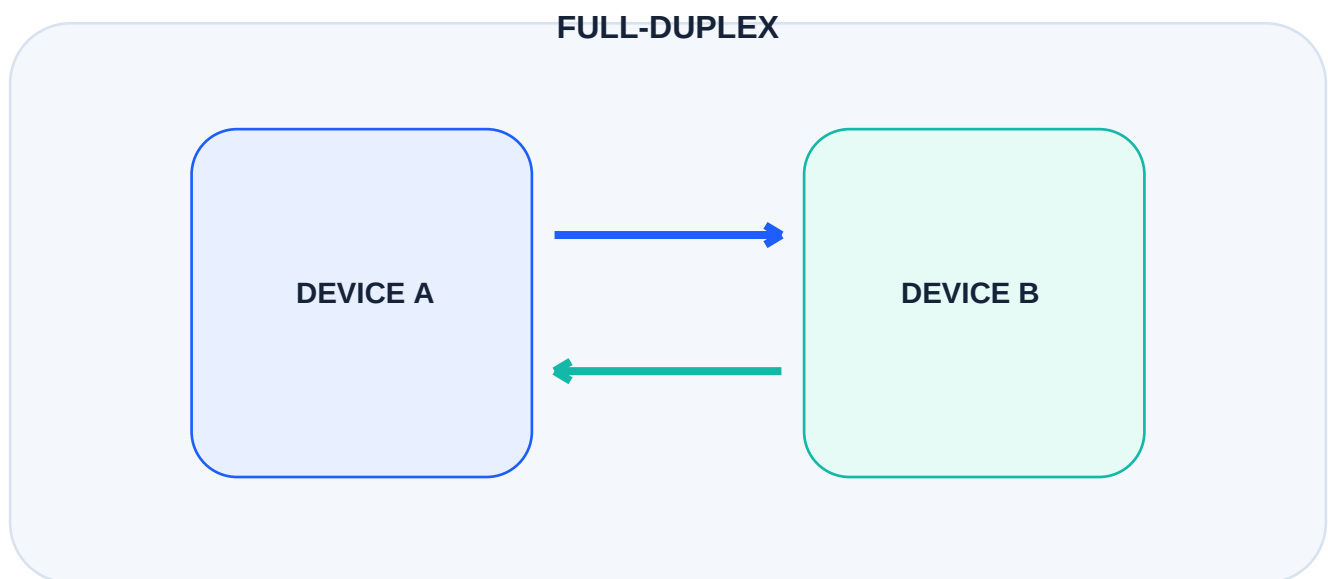
ANSWER

4 (=8 wires)

10GBASE-T uses all four pairs.

QUESTION

A [...] connection allows both devices to send and receive data at the same time.



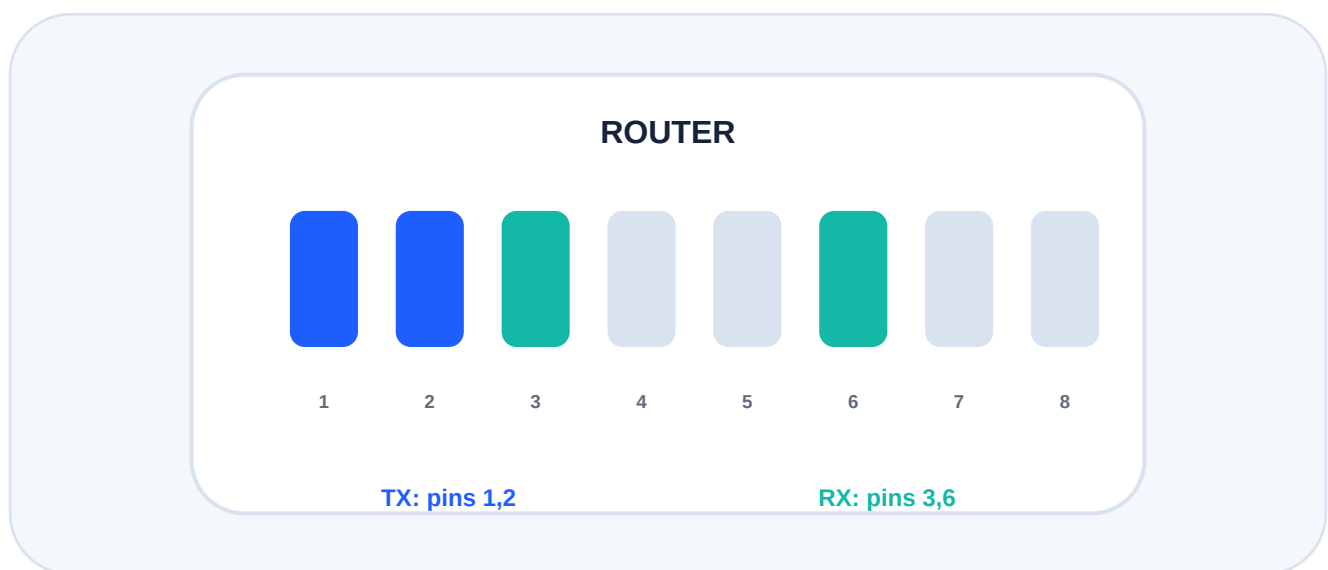
ANSWER

full-duplex

Full-duplex means simultaneous sending and receiving.

QUESTION

In a FastEthernet connection, a router transmits data on which pins [...]



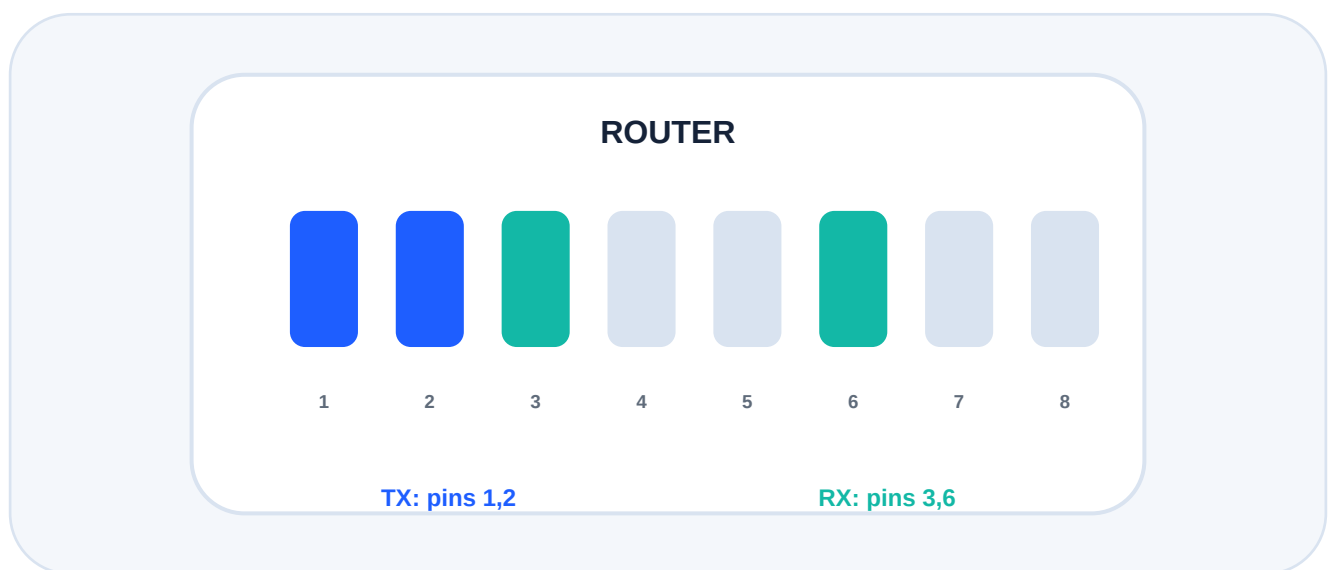
ANSWER

1 and 2

Routers use pins 1-2 to transmit in 10/100BASE-T.

QUESTION

In a FastEthernet connection, a router receives data on which pins [...]



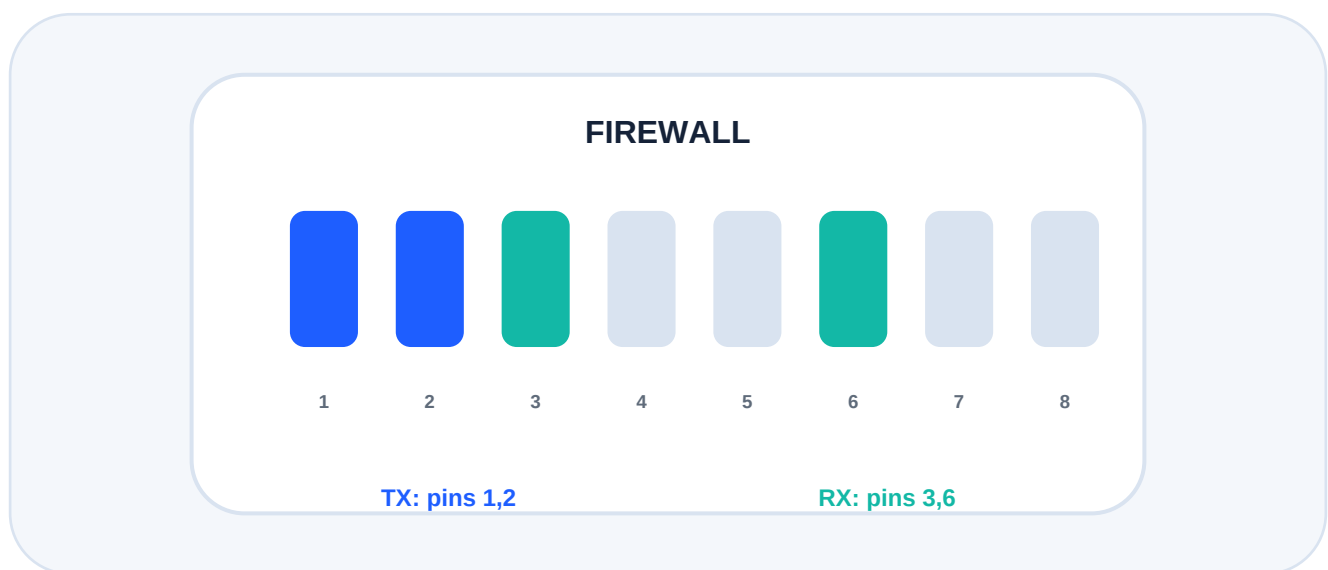
ANSWER

3 and 6

Routers use pins 3-6 to receive in 10/100BASE-T.

QUESTION

In a FastEthernet connection, a firewall transmits data on which pins [...]



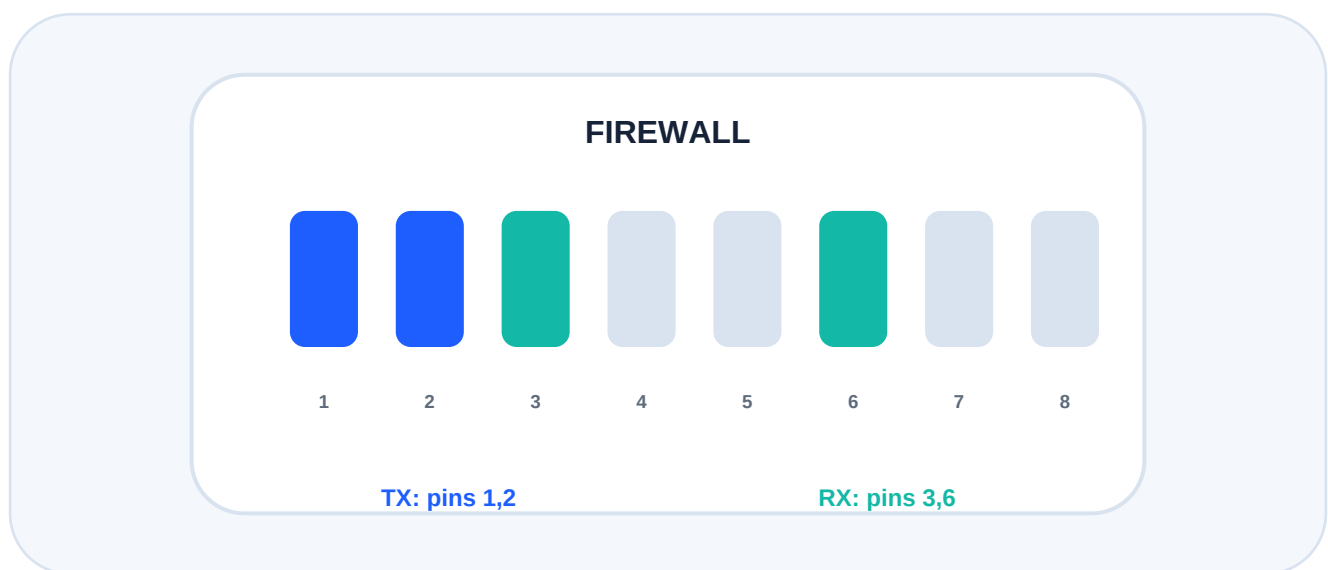
ANSWER

1 and 2

Firewalls follow the same MDI pin behavior as routers/PCs.

QUESTION

In a FastEthernet connection, a firewall receives data on which pins [...]



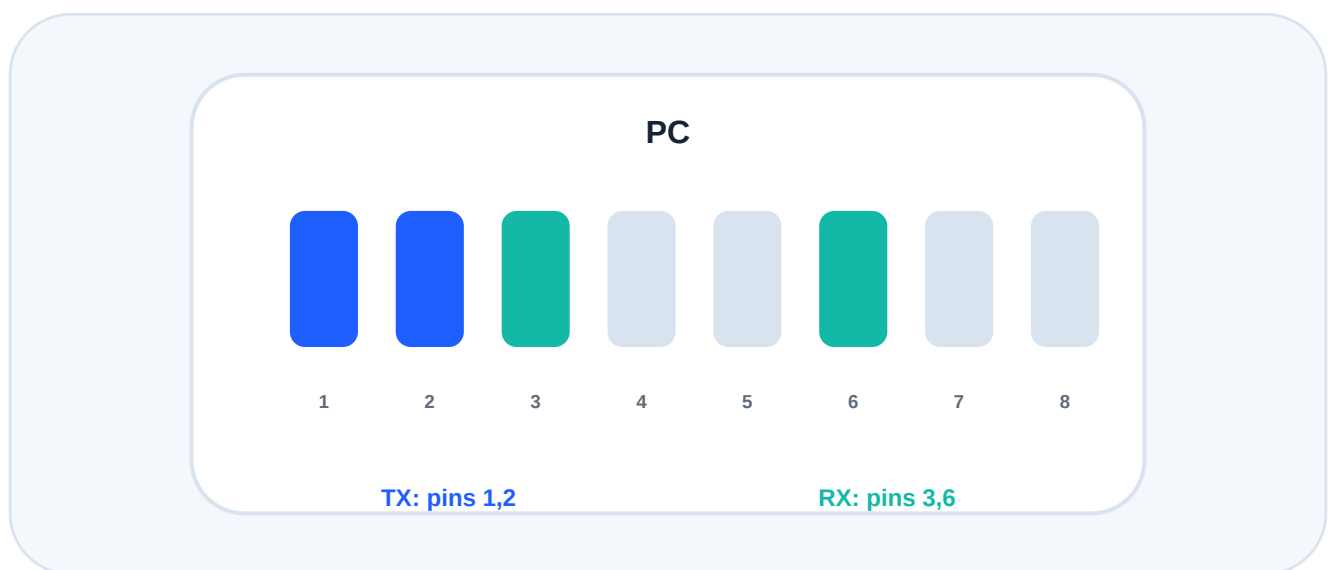
ANSWER

3 and 6

Firewalls receive on pins 3-6 in 10/100BASE-T.

QUESTION

In a FastEthernet connection, a PC transmits data on which pins [...]



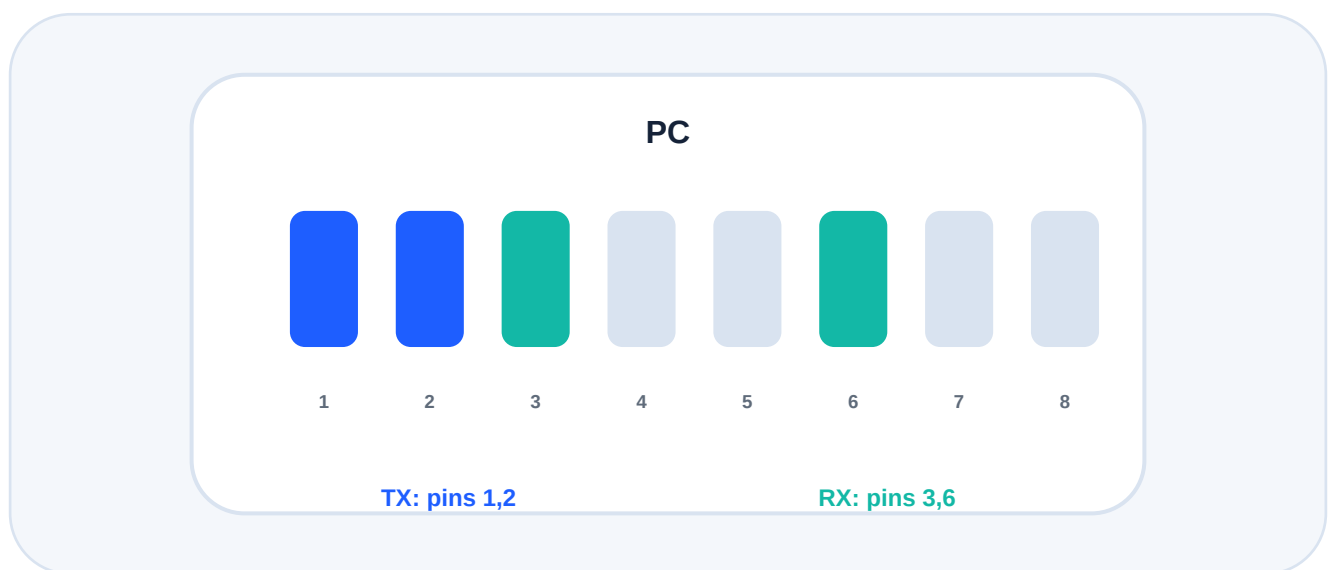
ANSWER

1 and 2

PCs transmit on pins 1-2 in 10/100BASE-T.

QUESTION

In a FastEthernet connection, a PC receives data on which pins [...]



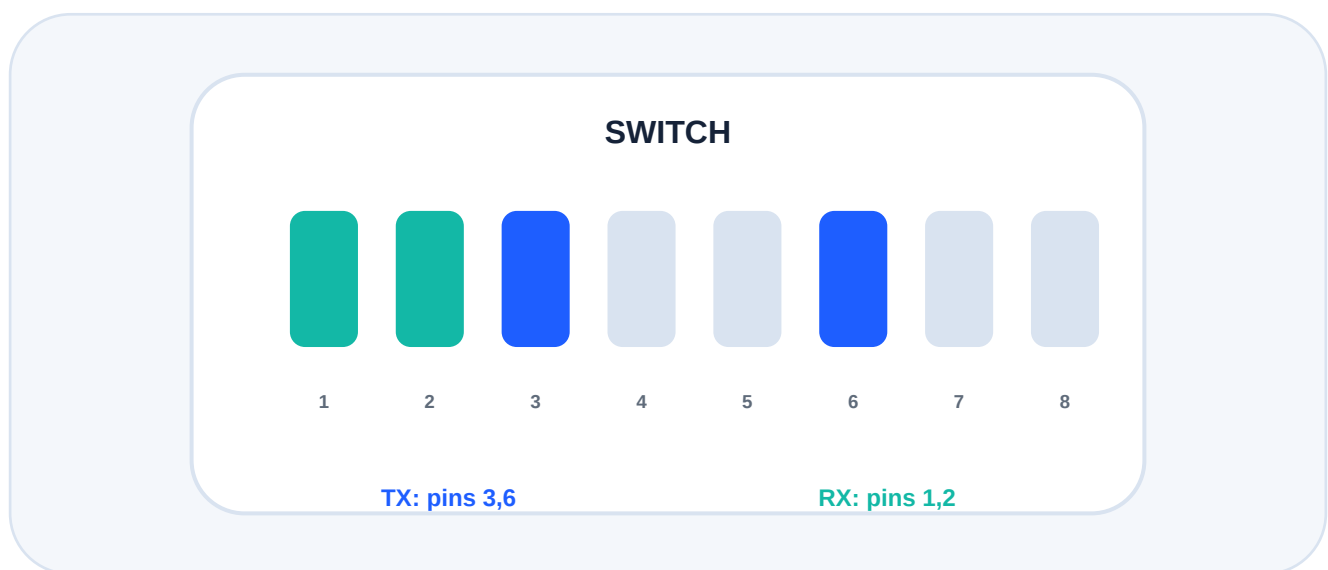
ANSWER

3 and 6

PCs receive on pins 3-6 in 10/100BASE-T.

QUESTION

In a FastEthernet connection, a switch transmits data on which pins [...]



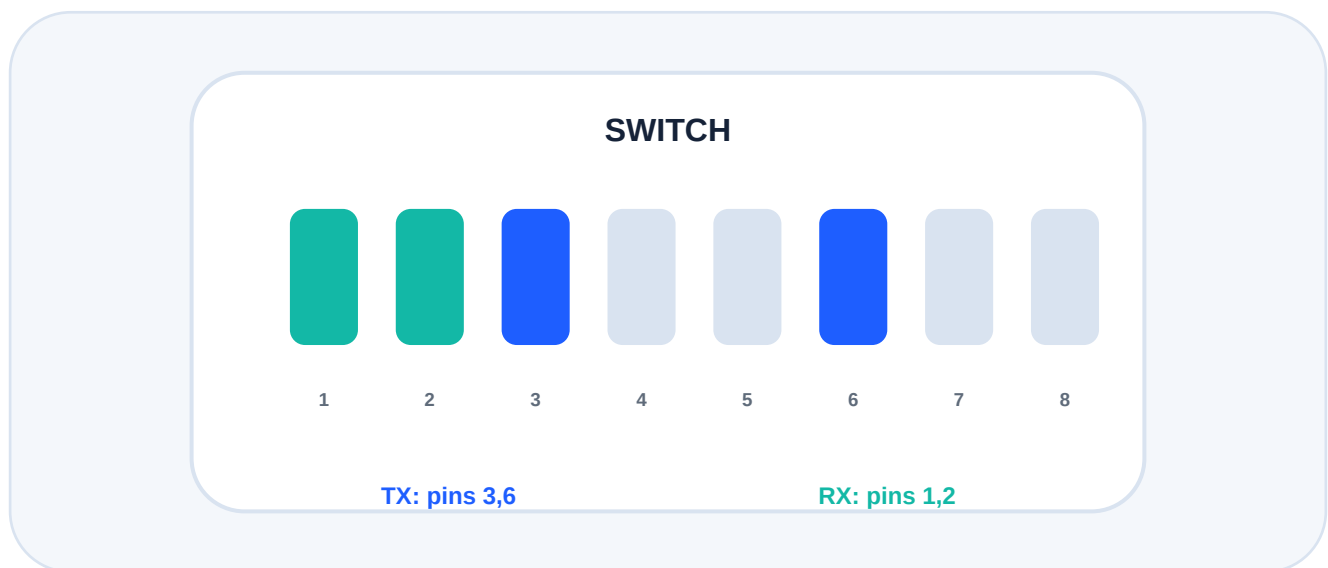
ANSWER

3 and 6

Switch ports transmit on pins 3-6 in 10/100BASE-T.

QUESTION

In a FastEthernet connection, a switch receives data on which pins [...]



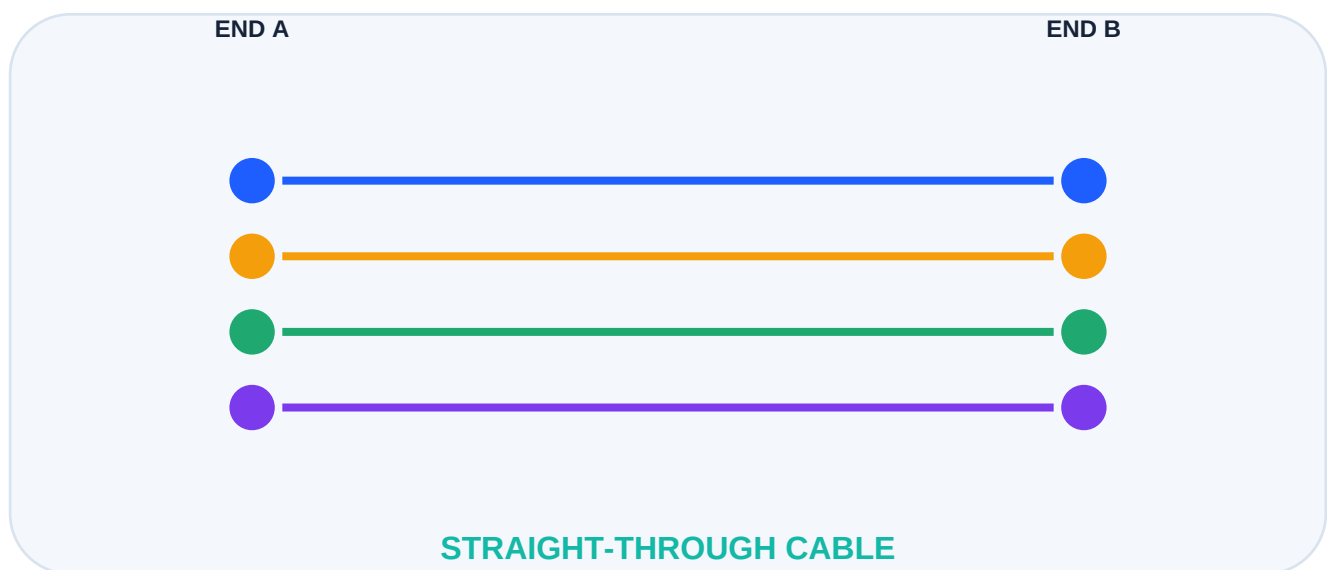
ANSWER

1 and 2

Switch ports receive on pins 1-2 in 10/100BASE-T.

QUESTION

If Auto MDI-X is disabled, what kind of UTP cable should be used to connect a router and a switch?



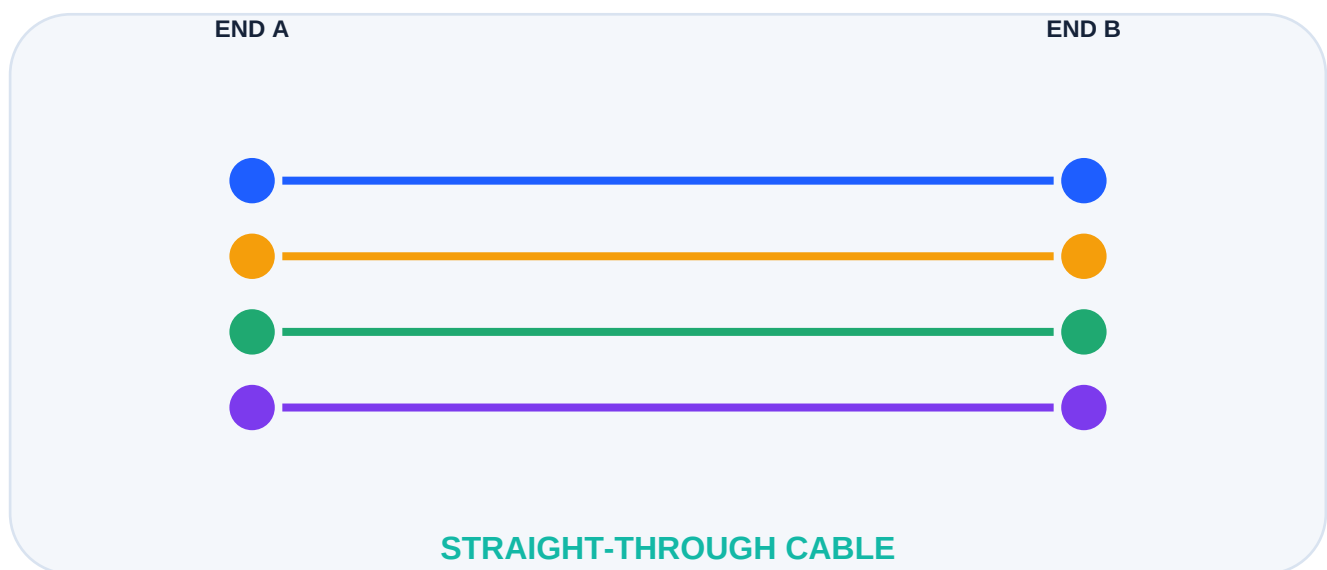
ANSWER

straight-through cable

Router-to-switch uses straight-through when Auto MDI-X is off.

QUESTION

If Auto MDI-X is disabled, what kind of UTP cable should be used to connect a PC and a switch?



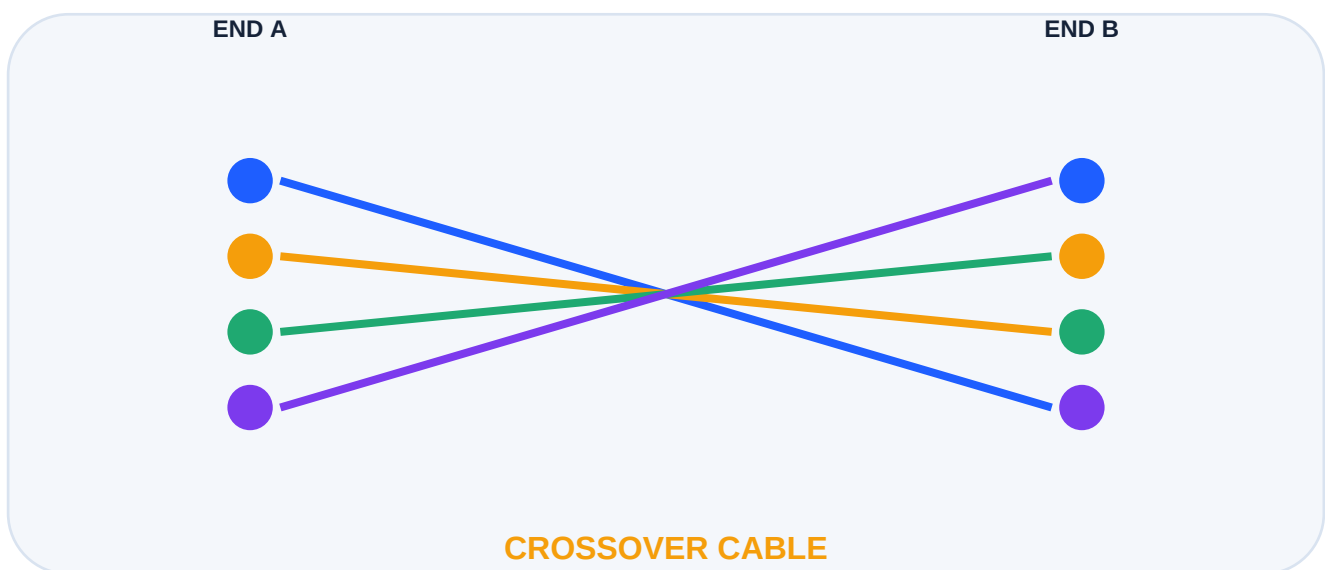
ANSWER

straight-through cable

PC-to-switch uses straight-through when Auto MDI-X is off.

QUESTION

If Auto MDI-X is disabled, what kind of UTP cable should be used to connect two routers together?



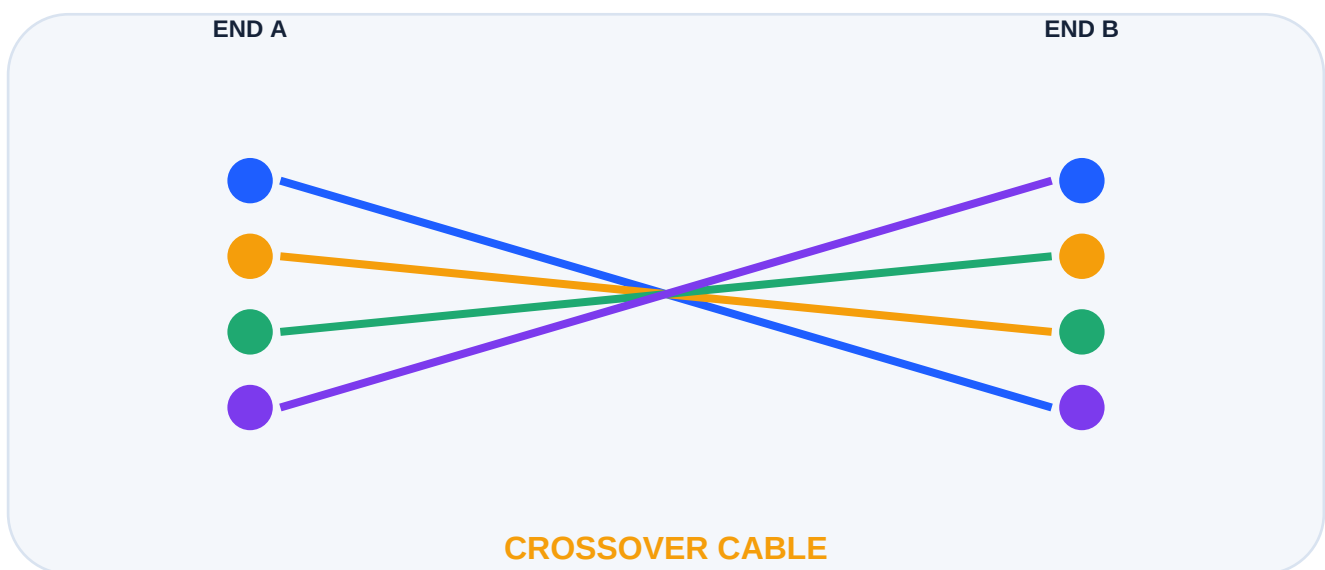
ANSWER

crossover cable

Like MDI devices need crossover when Auto MDI-X is off.

QUESTION

If Auto MDI-X is disabled, what kind of UTP cable should be used to connect two switches together?



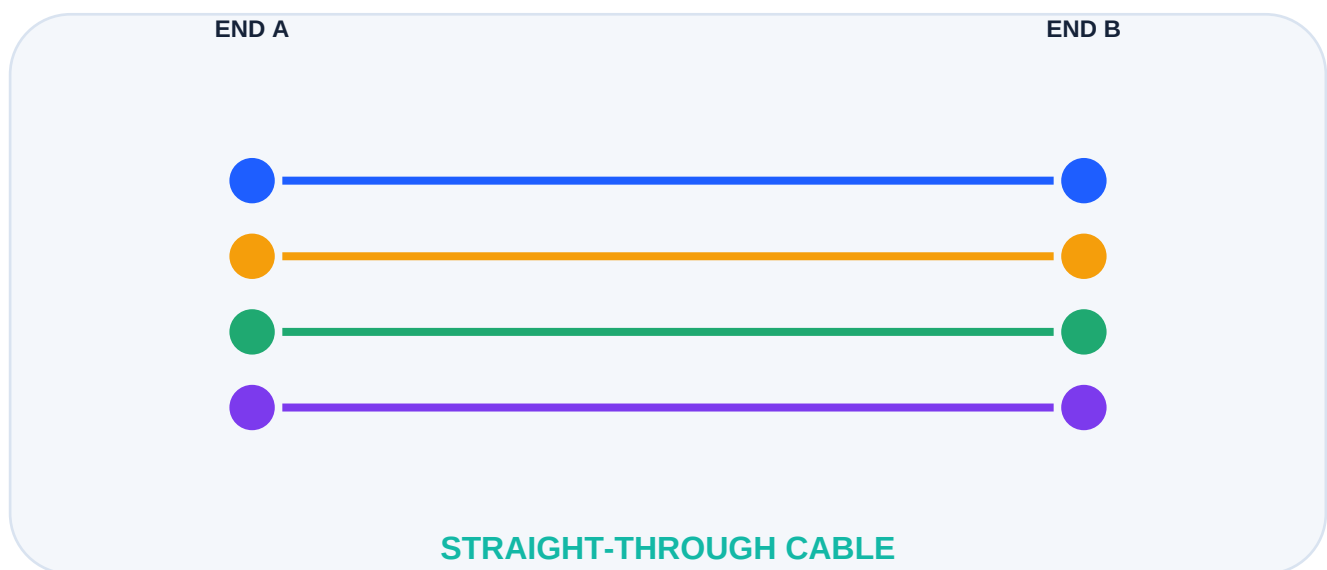
ANSWER

crossover cable

Like switch ports need crossover when Auto MDI-X is off.

QUESTION

What kind of cable connects a pin pair on one end of a UTP cable to the same pair on the other end?



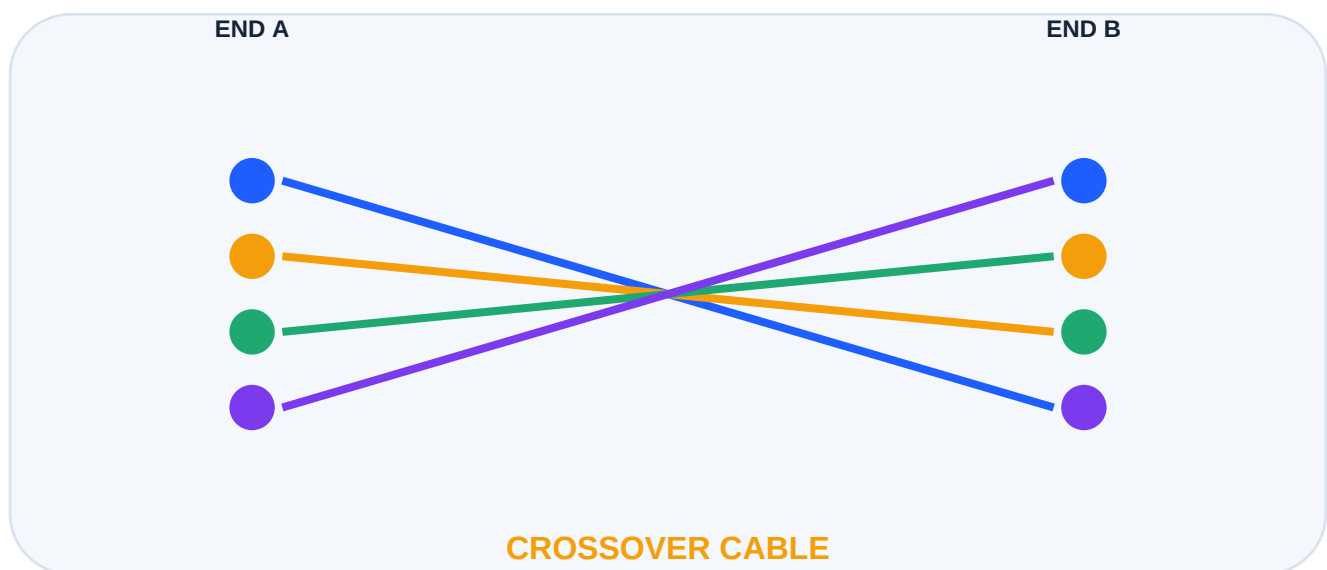
ANSWER

straight-through cable

Straight-through preserves the same pair mapping end-to-end.

QUESTION

What kind of cable connects a pin pair on one end of a UTP cable to the opposite pair on the other end?



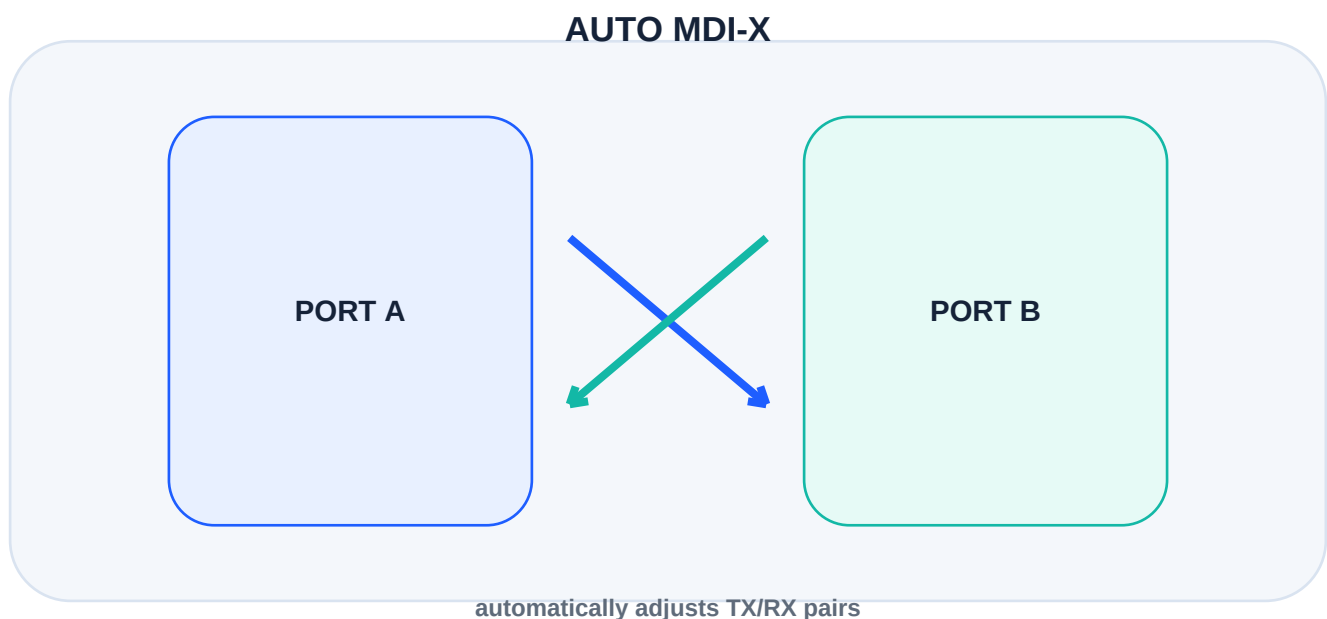
ANSWER

crossover cable

Crossover swaps transmit and receive pairs.

QUESTION

What is the name of the feature that allows a device to automatically adjust which RJ45 pin pairs it uses to transmit and receive data?



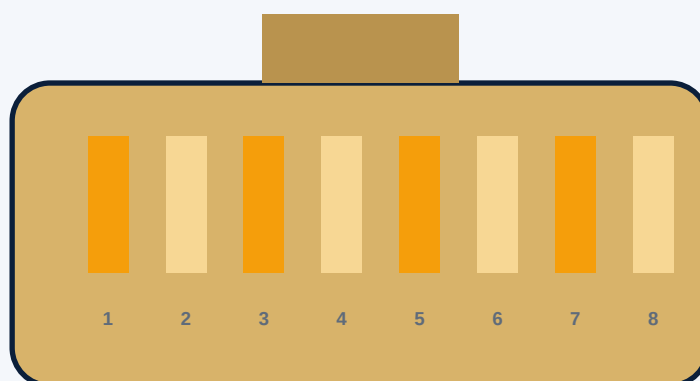
ANSWER

Auto MDI-X

Auto MDI-X automatically corrects TX/RX pair selection.

QUESTION

Which two pin pairs are used in a 10BASE-T or 100BASE-T connection?



RJ45 CONNECTOR

Used pairs: 1-2 and 3-6

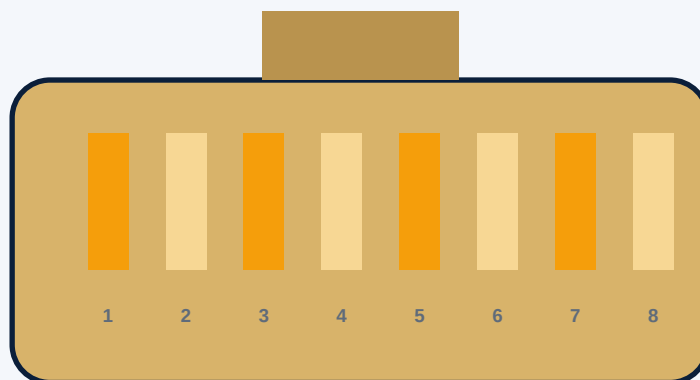
ANSWER

Pair 1-2, Pair 3-6

10/100BASE-T uses pairs 1-2 and 3-6.

QUESTION

Which four pin pairs are used in a 1000BASE-T or 10GBASE-T connection?



RJ45 CONNECTOR

Used pairs: 1-2, 3-6, 4-5, 7-8

ANSWER

Pair 1-2, Pair 3-6, Pair 4-5, pair 7-8

Gigabit and 10GBASE-T use all four pairs.

QUESTION

Fiber-Optic cables are connected to [...] transceivers in a router or switch.

SFP TRANSCEIVER

Small Form-Factor Pluggable

ANSWER

SFP (Small Form-Factor Pluggable)

Fiber links commonly plug into SFP-family transceivers.

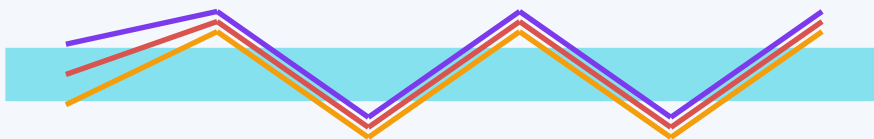
QUESTION

What are the two main kinds of fiber-optic cable?

SINGLE-MODE



MULTIMODE



ANSWER

Single-mode and multimode

The main fiber categories are single-mode and multimode.

QUESTION

Multimode Fiber cables have a [...] fiberglass core than single-mode fiber cables.

MULTIMODE

wider core

SINGLE-MODE

narrower core

ANSWER

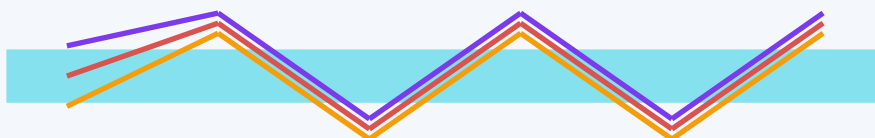
wider

Multimode has the wider core.

QUESTION

[...] fiber cables allow multiple angles (modes) of light waves to enter the fiberglass core.

MULTIMODE



ANSWER

Multimode

Multimode allows multiple light paths or modes.

QUESTION

Which allows longer cables, single-mode or multimode fiber?

SINGLE-MODE

longer reach

MULTIMODE

shorter reach

ANSWER

single-mode

Single-mode supports longer distances.

QUESTION

Which is cheaper, single-mode or multimode fiber?

MULTIMODE

cheaper

SINGLE-MODE

more costly

ANSWER

multimode

Multimode optics/cabling are typically cheaper for short links.

QUESTION

In [...] fiber cables, light travels straight through the fiberglass core at a single angle.

SINGLE-MODE



ANSWER

single-mode

Single-mode carries light along a single path.

QUESTION

10GBASE-SR, 10GBASE-LR, and 10GBASE-ER were defined in the IEEE [...] standard.

IEEE

802.3ae

10GBASE-SR / LR / ER

ANSWER

802.3ae

802.3ae defines major 10 Gigabit fiber variants.

QUESTION

1000BASE-LX was defined in the IEEE [...] standard.

IEEE

802.3z

1000BASE-LX

ANSWER

802.3z

802.3z includes 1000BASE-LX.

QUESTION

1000BASE-LX maximum cable length:



Multimode

550 meters



Single-mode

5 kilometers

ANSWER

550 meters (multimode)
5 kilometers (single-mode)

1000BASE-LX reaches farther on single-mode fiber.

QUESTION

10GBASE-SR maximum cable length:



10GBASE-SR

400 meters

ANSWER

400 meters

SR = short reach.

QUESTION

10GBASE-LR maximum cable length:



10GBASE-LR

10 kilometers

ANSWER

10 kilometers

LR = long reach.

QUESTION

10GBASE-ER maximum cable length:



10GBASE-ER

30 kilometers

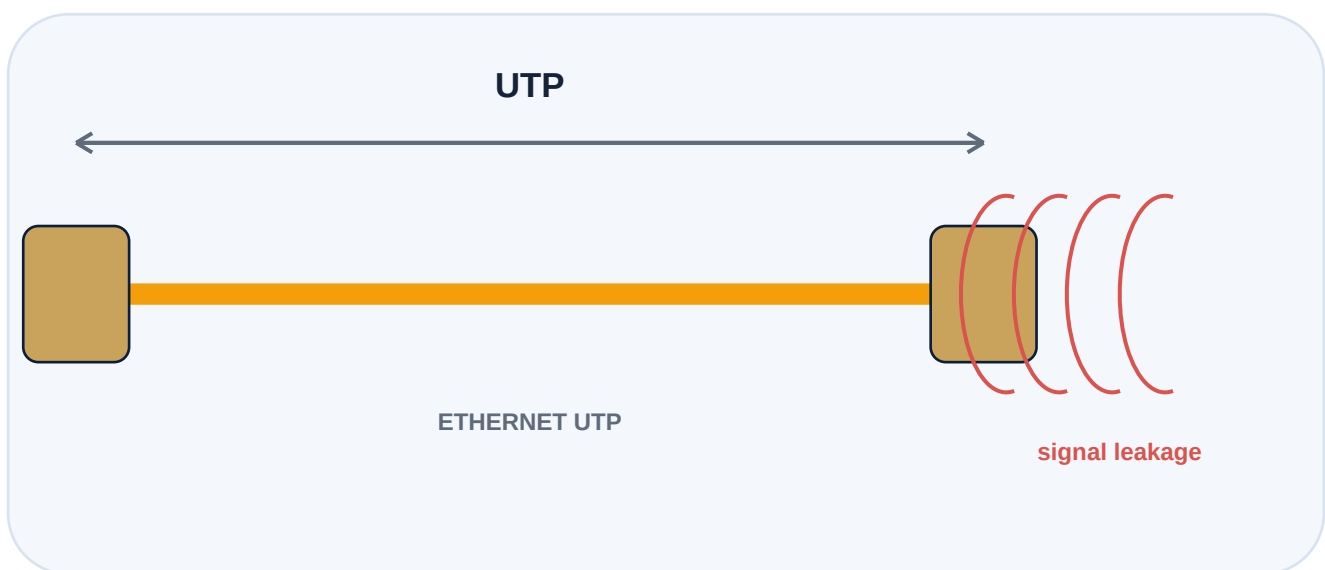
ANSWER

30 kilometers

ER = extended reach.

QUESTION

[...] cables emit a faint signal outside of the cable, which can be detected and copied.



ANSWER

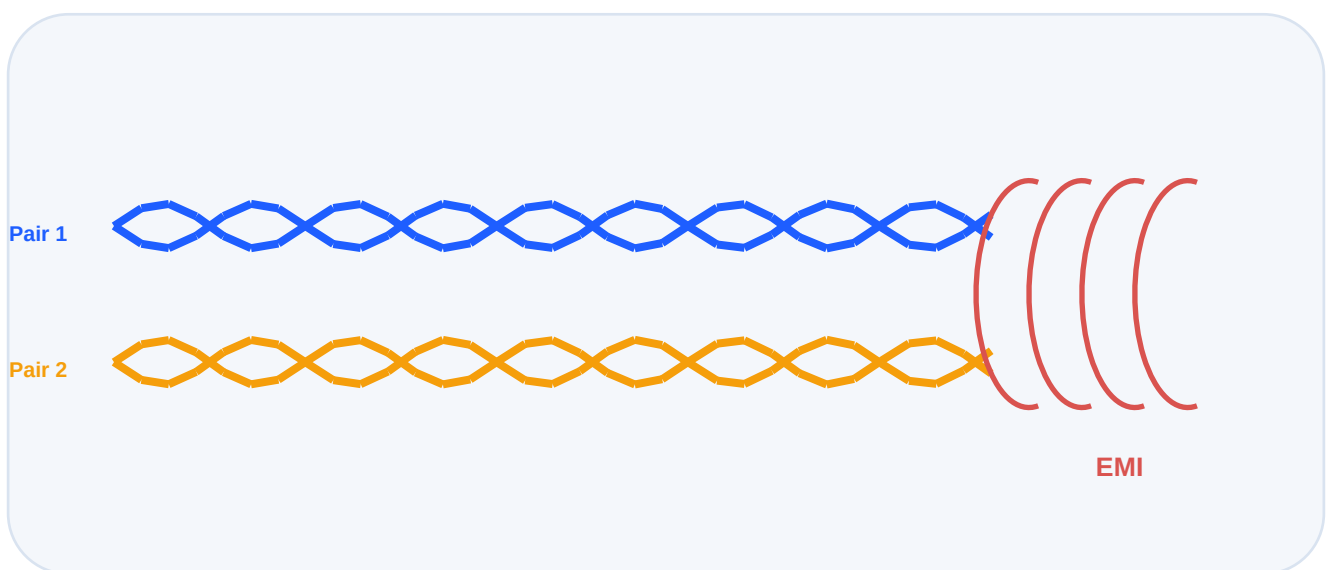
UTP

Copper UTP can radiate a small electromagnetic signal.

QUESTION

[...] cables are vulnerable to EMI (electromagnetic interference).

TWISTED PAIRS



ANSWER

UTP

Because UTP is unshielded, it is susceptible to EMI.

DAY 02 - QUICK REVIEW

COPPER ETHERNET

● Connector	RJ45 - 8 pins
● Max UTP length	100 meters
● UTP	Unshielded Twisted Pair
● 10/100BASE-T	2 pairs / 4 wires
● 1000/10GBASE-T	4 pairs / 8 wires

CABLE BEHAVIOR

● Full-duplex	Send + receive simultaneously
● Straight-through	Same pair mapping
● Crossover	TX/RX pairs crossed
● Auto MDI-X	Automatically adjusts TX/RX pairs

FIBER

● Transceiver	SFP
● Types	Single-mode + multimode
● Single-mode	Longer reach
● Multimode	Wider core; typically cheaper

KEY FIBER DISTANCES

● 1000BASE-LX	550 m MM / 5 km SM
● 10GBASE-SR	400 m
● 10GBASE-LR	10 km
● 10GBASE-ER	30 km

Fast pin rule for 10/100BASE-T

Router / Firewall / PC: TX 1-2, RX 3-6 | Switch: TX 3-6, RX 1-2